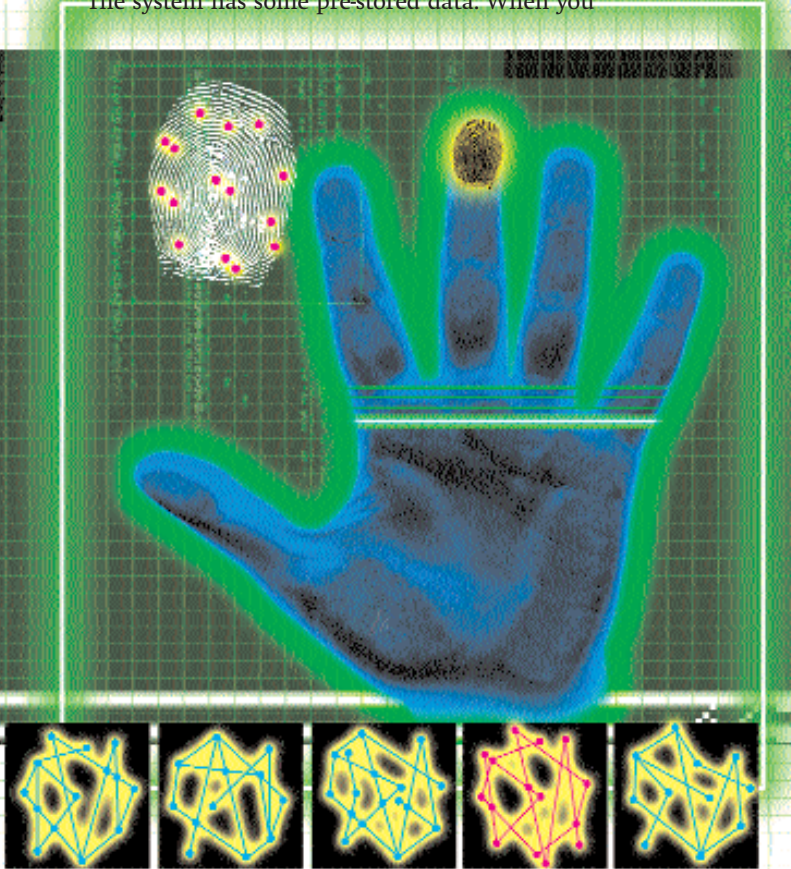


and anyone with basic knowledge of computers can crack passwords using the myriad free programmes available on the Internet.

For obvious reasons, a fingerprint, however, is difficult to fake without the help of the owner. It is also a lot more convenient to simply place your finger on to a scanner instead of remembering a long and complex series of characters and their cases. To make matters worse, they should, ideally, have no correlation whatsoever. So essentially, for maximum security, your password should be an alphanumeric 'word' that doesn't mean anything. Go figure!

How Biometric Processes Work

The concept of Biometric verification is simple. The system has some pre-stored data. When you



approach the system (say a fingerprint scanner), your finger is scanned and matched with a record of fingerprints already in its database. Only when it finds a match, access is granted. The concept might be simple, but the process is quite ingenious.

Fingerprint Scanning

Humans have fingerprints for the exact same reason that tyres have treads. It helps in better grip and, by a bizarre twist of nature, different people have entirely different sets of fingerprints, which enables identification.

A fingerprint is made up of ridges and valleys (lines and the gaps separating them) and it is these ridges and valleys which are scanned to verify the authenticity of a print.

To authenticate a set of prints, a scanner needs to do two things: first, it needs to get the



Each fingerprint has a unique pattern of ridges and valleys

image of the prints which are to be authenticated, and second, it needs to actually go about the business of verifying them.

The most commonly used method of scanning is optical scanning. An optical scanner has a CCD sensor (Charge Coupled Device) similar to the ones used in digital cameras. There is an array of light sensitive diodes (photosites). When these diodes come in contact with light, they generate an electrical signal. Every photosite records a pixel representing the light it came in contact with.

An analogue to digital converter (ADC) system in the scanner processes the electrical signals to generate digital representations of the image. It is not necessary that the same kind of light falls on all diodes. So what is generated is a mix of dark and light areas, which together make up the image.

The process begins as soon as you place your finger on the glass plate. The scanner has its own source of light (mostly an array of LEDs) which illuminate the finger and the CCD inside takes a picture of the finger. But it doesn't just rush off to match it with the images in the database. It first checks for the integrity of the image in terms of contrast, sharpness and sheer quality.

The system checks the average pixel darkness, or, might employ a sampling technique and check the overall values in a small sample area. If the image is too dark or too light, it is rejected. Exposure settings are then accordingly adjusted and the print rescanned.

If the exposure level is found to be correct, it goes on to check the image definition (sharpness of the fingerprint). It does so by analysing several straight lines moving horizontally and vertically across the image. If the definition is good, a line running perpendicular to the ridges will comprise alternating segments of light and dark pixels. If this is found to be in order, the scanner proceeds to comparing the captured fingerprint with those in the database to see if it can find a match.

Matching of prints is a fairly complex process in itself and is far removed from the super-imposing method commonly shown in films.

This is so because smudging (due to scan surface or oily fingers) can make the same print appear different in different photos. Also, scanning and matching the entire finger consumes a lot of processing power.

Instead, the scanners compare specific features of the fingerprint (called minutiae). These points are generally places where ridge lines end or bifurcation occurs. The idea is to